



Hoof lesion detection in lactating Holsteins: Part I. External validation of autonomous camera-derived locomotion score thresholds

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ABSTRACT

Hoof lesion detection remains a challenge in lameness management. Recent studies have proposed locomotion score-based thresholds using an autonomous camera system (AUTO), which generates locomotion scores (LS) ranging from 0 (sound) to 100 (severely lame). However, these thresholds have been developed and evaluated within the same study populations, and their performance on independent herds remains unknown, raising concerns about generalizability. The primary objective of this study was to evaluate the classification performance of AUTO-derived locomotion metrics for identifying cows with hoof lesions. Specifically, we assessed the performance of a weekly average LS threshold ≥ 50 (50_BIN) and 4 monthly AUTO-derived locomotion metrics: average LS (mAVG), minimum LS (mMIN), maximum LS (mMAX), and percentage of LS scores above 50 (mPLS). Performance metrics, including positive predictive value (PPV) and negative predictive value (NPV), were calculated for each threshold. Holstein cows ($n = 511$) of 8 to 100 DIM, with no active lesions and no more than 1 prior noninfectious lesion, were enrolled from 2 sites. Cows were subsequently classified as typically treated, which included cases of digital dermatitis, sole ulcer, toe lesion, white line disease, or other hoof lesions requiring treatment; or as typically not treated, which included cows without hoof lesions or without lesions typically requiring treatment. For positive classifications at the overall parity level, 50_BIN achieved the highest PPV ($n = 451$; PPV = 25.3%, 95% CI: 16.1–35.1), whereas mPLS achieved the highest NPV ($n = 389$; NPV = 92.4%, 95% CI: 89.0–95.6). These results highlight that threshold-based approaches failed to reliably identify cows with lesions, mainly due to low PPV. Based on these findings, we do not support the on-farm adoption of these AUTO-derived threshold methods for the identification and

differentiation of cows with hoof lesions. The low PPV indicates that adopting these thresholds could increase unnecessary hoof trimming, associated labor demands, and economic costs. Future research should prioritize improving PPV to accurately identify cows with lesions and avoid potentially unnecessary hoof trimming visits.

Key words: lameness, dairy cow, precision livestock farming

INTRODUCTION

Lameness remains a challenge in the dairy industry due to its negative impact on animal welfare (Whay and Shearer, 2017), economic losses (Cha et al., 2010; Ózsvári, 2017), and increased greenhouse gas emissions (Mostert et al., 2018). Importantly, lameness is not a disease but a clinical indicator of impaired locomotion (Archer et al., 2010). In dairy cattle, this impaired locomotion is most often associated with underlying hoof lesions, and cows with hoof lesions are more likely to exhibit lameness (van Huyssteen et al., 2020). Therefore, effective lameness mitigation depends on timely and accurate identification of cows with clinically relevant hoof lesions.

On most dairy farms, lameness detection relies on passive surveillance. Although regular visual locomotion scoring has been shown to improve lameness management (Leach et al., 2012; Thomas et al., 2015; Wilson et al., 2022), it is time- and labor-intensive and prone to human bias, including underestimation of lameness prevalence when only severely lame cows are recognized as lame (Leach et al., 2010). These limitations have motivated interest in automated technologies capable of providing continuous, objective locomotion scores at the farm level.

Autonomous camera-based systems (AUTO) have emerged as a promising alternative for locomotion monitoring. A commercially available system (CattleEye Ltd., Belfast, UK) has been shown to perform comparably to human assessors in locomotion scoring and has been proposed as a tool for identifying cows that may ben-

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

efit from hoof trimming (Anagnostopoulos et al., 2023). When applying a weekly locomotion score threshold of 50 or greater, Anagnostopoulos et al. (2023) reported that AUTO outperformed one human assessor in detecting cows with potentially painful hoof lesions. However, Swartz et al. (2024b) observed substantial overlap in locomotion score distributions between cows with and without lesions, limiting the effectiveness of a single-threshold approach. Building on these findings, Siachos et al. (2025) proposed additional threshold-based metrics derived from monthly summaries of AUTO locomotion scores (i.e., monthly average, minimum, maximum, and percentage of scores above 50) to improve detection of moderate to severe hoof lesions.

Despite these advances, a critical limitation remains: previously proposed threshold-based approaches have been developed and evaluated largely within the same study populations, and their performance has not been externally validated across independent herds. Thresholds developed within the same study population risk overfitting the underlying locomotion score distribution and perform differently when applied to new herds with distinct cow cohorts, management conditions, or data characteristics (J. Gu, University of Munich, Munich, Germany, and Siemens AG, Corporate Technology, Munich, Germany; D. Oelke, Siemens AG, Corporate Technology, Munich, Germany; unpublished data). Without external validation, reported diagnostic performance may be overly optimistic and may not generalize to different management systems or populations (Bleeker et al., 2003). Unlike previous evaluations, our study focuses on identifying cows that require therapeutic intervention, thereby providing a more clinically meaningful test of threshold performance. To address this concern, this study aimed to externally validate diagnostic test characteristics of the locomotion score threshold of 50+ proposed by Anagnostopoulos et al. (2023), as well as the additional threshold-based recommendations introduced by Siachos et al. (2025) to detect cows with a hoof lesion requiring treatment.

MATERIALS AND METHODS

Study Design and Housing and Cow Characteristics

The data for this study came from a case-cohort study (O'Brien et al., 2022) conducted from July to November 2023 on a conventional, commercial dairy operation located in Minnesota, consisting of 2 sites that differed in facility design based on parity. The dairy operation was selected due to a prior working relationship and the existing installation of the AUTO CattleEye system. Both sites used freestall housing with grooved concrete flooring and followed a milking schedule of 3 times per

day. Cows were housed indoors throughout their lactation period. Site 1 housed multiparous cows in a freestall facility equipped with a combination of deep-bedded stalls and mattresses topped with manure solids. Site 2 was designated for primiparous cows and featured a similar freestall system with mattresses bedded with manure solids.

All cows calved at site 1. First-lactation cows were transferred to site 2 at 10 ± 3 DIM, unless a health event delayed their movement. No a priori sample size calculation was conducted for this study; the current analysis uses the same cohort of cows enrolled for the machine learning-focused study (Swartz et al., 2026), whose design was informed by funding constraints and sponsor guidance (CattleEye Ltd., Belfast, UK). All protocols involving animals in this study were approved by the University of Minnesota Institutional Animal Care and Use Committee (St. Paul, MN), protocol number 2206-40148A.

Enrollment and Follow-Up Evaluation

Cows were eligible for this study if they were Holstein dairy cows between 8 and 100 DIM, had left the transition pen, and had no more than 1 noninfectious hoof lesion before enrollment. Each cow underwent an enrollment evaluation with eligible cows that remained in the herd undergoing a follow-up evaluation occurring 100 to 120 d later. For each evaluation, the animals were brought to a hoof trimming chute, where their radio frequency identification (RFID) was collected using an RFID reader (AWR300 Stick Reader, Allflex) and cross-verified with the operation's management software (Dairy Comp 305, Valley Agricultural Software, Tulare, CA) to identify any discrepancies in RFID registration linked to cow identification numbers.

For the enrollment evaluation, each cow received a superficial trim (removal of a superficial sliver of horn) performed by 1 of 2 study personnel to facilitate lesion inspection while keeping the hoof as close as possible to its untreated condition. For the follow-up evaluation, each cow received a routine hoof trim using the modified functional method with a larger model on the weight-bearing hoof (Stoddard et al., 2025). Following the hoof trimmings for each evaluation, a visual hoof inspection was conducted using the 13-zone system (Shearer et al., 2004), and findings were recorded. Hock and knee evaluations were also performed, with lesions recorded as present if ulcerative. Before the start of the study, author GC trained author DS in hoof lesion recording, and this training continued into the early phases of the study. Hoof lesion recordings were conducted by both GC and DS, with the majority performed by DS. However, whenever DS expressed uncertainty, GC provided clarification

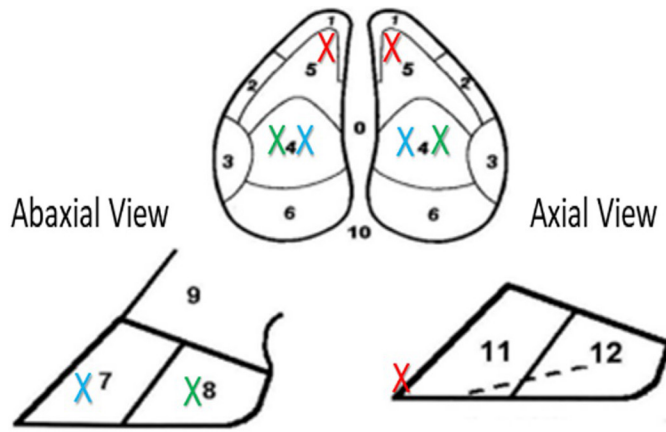


Figure 1. Illustration of the 13-zone hoof recording system adapted from Shearer et al. (2004). Hoof testers were applied to assess pain sensitivity by compressing matched zones simultaneously. Zones marked with red X's, blue X's, and green X's were each compressed as pairs.

to ensure consistency. In addition to the visual assessment, hoof testers were applied 3 times to each hoof, with each application targeting a different zone to assess for a withdrawal reflex (Figure 1). If there was a positive withdrawal response, the hoof testers were reapplied to

ensure a consistent response. When positive withdrawal reflexes were present without visible hoof lesions, GC conducted a more thorough inspection of the hoof to ensure no hoof lesions were present.

All data were recorded in a computerized lesion recording system using a rugged touchscreen tablet (Hoof Supervisor, Dresser, WI). At enrollment, cows with digital dermatitis in the interdigital cleft presenting as an ulcerative lesion ≥ 2 cm in diameter (stage M2), swelling of the first phalanx (P1) or second phalanx (P2), hock or knee lesions with swelling (lesions without swelling were eligible to continue), sole ulcer, toe lesion, or white line disease, or those that elicited a withdrawal reflex, were treated and excluded from the study. Descriptions for these lesions are provided in Table 1.

Cow and Hoof Trimming Data

Cow-level data were extracted from the operation's herd management software. The data included individual cow lactation number and hoof trimming history. Hoof trimming history included records of trimmings performed by the site-specific hoof trimmers ($n = 2$), both of whom had been previously trained and calibrated

Table 1. Identified conditions, description of each condition, study eligibility, and classification group designations as lesion not requiring treatment (LNRT), lesion requiring treatment (LRT), or pain (adapted from Claw Health Atlas, Egger-Danner et al., 2020)

Diagnosis	Zone ¹	Definition	Enrollment eligible ²	Outcome classification
Corkscrew claw	7, 8	Any torsion of either the outer or inner hoof; the dorsal edge of the wall deviates from a straight line	Yes	LNRT
Heel erosion	6	Eroded heel horn that involved a flap or severely degraded horn tissue	Yes	LNRT
Horizontal fissure	7, 8	Horizontal crack in the hoof wall	Yes	LNRT
Interdigital hyperplasia	0	Interdigital growth of fibrous tissue	Yes	LNRT
M1 ³	10	An ulcerative lesion less than 2 cm in diameter	Yes	LNRT
M2 ³	10	An ulcerative lesion greater than or equal to 2 cm in diameter	No	LRT
M3 ³	10	A scab-like appearance covering where a M1 or M2 would have been located	Yes	LNRT
M4 ³	10	Raised skin with hyperkeratotic or proliferative growths	Yes	LNRT
M4.1 ³	10	An ulcerative lesion that occurs on a M4c growth that is less than 2 cm in diameter	Yes	LNRT
Swollen	Non-hoof ⁴	Any swelling occurring on the P1, P2, or hock	No	LNRT
Hock/knee	Non-hoof ⁴	Any ulcerative and bleeding lesions occurring on the hock or knee that are not swollen	Yes	LNRT
Other ⁵	Unknown ⁶	Other outcomes recorded by the site hoof trimmers:	NA ⁷	LRT
Sole hemorrhage	4	Diffused and/or circumscribed red or yellow discoloration of the sole	Yes	LNRT
Sole ulcer	4	A defect in sole horn that exposes corium	No	LRT
Toe lesion	1, 2, 5	Combination of thin soles (horn that yields with digital pressure) and toe ulcer (exposure of corium through the sole horn)	No	LRT
White line hemorrhage	3	Diffused and/or circumscribed red or yellow discoloration of the white line	Yes	LNRT
White line disease	3	Separation of the white line with or without purulent exudation	No	LRT
Pain	Unknown ⁶	Any cow eliciting a withdrawal reflex with hoof testers	No	Pain

¹Zone in which the diagnosis occurred, according to the 13-zone scoring system (Shearer et al., 2004).

²Eligibility to continue if this diagnosis was found at the enrollment trimming.

³Stages of digital dermatitis (Berry et al., 2010).

⁴Diagnosis location not on the hoof.

⁵Different lesions reported by the site hoof trimmers: interdigital hyperplasia, periople ulcer, non-ulcer, other, puncture, vertical fissure, axial fissure, corkscrew.

⁶No reported zone for diagnosis.

⁷NA = not applicable, because the site hoof trimmers did not perform the enrollment evaluations.

by study personnel (GC) in lesion identification and recording.

Locomotion Camera System Data

In addition to operation-recorded data, data were collected using AUTO, which accurately identifies individual cows (Swartz et al., 2024a) and generates locomotion scores (LS) (Anagnostopoulos et al., 2023). Briefly, the system uses an overhead-mounted 2-dimensional camera (~4 m high) installed in the return alley of the milking parlor to capture video footage of each cow. Identification is achieved via RFID ear or collar tags and visual recognition, with recordings captured as cows walk 3 to 4 m beneath the camera. The cameras were set to operate during low-stall maintenance or cow sorting periods, to minimize disruptions in cow flow and to ensure each cow passed under a camera at least once daily. Specifically, for site 1, the camera was set up to operate from 1700 to 0030 h, and from 1300 to 2100 h for site 2. Daily video recordings are transmitted to Amazon Web Services, where AUTO algorithms assign LS on a scale from 0 to 100 in 1-point increments. The study personnel received LS from cows enrolled in the study directly from the AUTO company personnel at the conclusion of the study. At the data transfer point, personnel were blinded to the results of the follow-up evaluations.

Data Management and Analysis. All data processing and modeling were performed in R (v4.4.2, R Core Team, 2024) using RStudio (v.2024.9.1.394, Posit Team, 2024).

Creating Outcomes of Interest. Hoof lesion diagnoses at each cow's first trimming after enrollment, along with lesion definitions and initial classifications as lesion not requiring treatment (LNRT; including cows without hoof lesions or with lesions not requiring intervention beyond routine or preventative trimming), pain, or lesion requiring treatment (LRT; including cases requiring blocks or topical treatments), are detailed in Table 1. These initial categories were used to create the outcome of interest (classification outcome) for this study. Although the original intent was for all outcome classifications to occur during the follow-up evaluation, some cows were examined by the site hoof trimmer before that evaluation. In these cases, the classification outcome was determined based on the earliest available hoof trimming record after the enrollment evaluation.

Cows were classified into 2 outcome categories: typically not treated (TNT), which included cows with LNRT, regardless of pain status; and typically treated (TT), which included cows with 1 or more LRT. This outcome was selected for this comparison because both prior studies focused exclusively on identifying cows with hoof lesions, without accounting for cows that exhibited only a withdrawal reflex indicating pain.

Thresholds Evaluated. To evaluate the performance of prior thresholds on an external dataset, we applied the 7-d threshold from Anagnostopoulos et al. (2023) and the 30-d moderate-to-severe lesion thresholds from Siachos et al. (2025) to our full dataset; severe-only thresholds were not evaluated, because lesion severity was not recorded in our dataset.

For the threshold of Anagnostopoulos et al. (2023), we calculated the average LS score for d 1 to 7 before the date of the outcome, and if the value was 50 or greater (**50_BIN**), it was dichotomized as having a lesion or not. This comparison was limited to cows that had at least 1 score in that interval. This approach was similar to that used by Anagnostopoulos et al. (2023), who applied the threshold to the closest weekly average before their foot trimming session. To evaluate the overall and parity group-specific thresholds (parities 2, 3, and 4+) of Siachos et al. (2025), we evaluated the suggested AUTO-derived metrics over the 30 d preceding the outcome evaluation: average LS (mAVG), minimum LS (mMIN), maximum LS (mMAX), and percentage of LS scores above 50 (mPLS), as shown in Table 2. As in their study, this comparison was limited to cows with at least 10 recorded AUTO scores and at least 30 d between enrollment and the outcome evaluation date. Cows in parity 1 were excluded from the parity-specific analysis, as Siachos et al. (2025) did not provide corresponding threshold cutoffs for this group.

For all threshold-based approaches, we compared the lesion classifications derived from the listed thresholds against the observed outcomes at the follow-up evaluation, which served as the gold standard for lesion status. Performance was assessed using accuracy, positive predictive value (PPV), negative predictive value (NPV), sensitivity, and specificity. Additionally, true prevalence

Table 2. Suggested thresholds for AUTO locomotion score metrics from Anagnostopoulos et al. (2023) and Siachos et al. (2025), stratified by parity group; for each metric, cows were classified as having a hoof lesion if their value was greater than or equal to the corresponding threshold

Group	mAVG ¹	mMAX ²	mMIN ³	mPLS ⁴	50_BIN ⁵
Overall	44.5	58.5	32.5	21.2	50
Parity 2	43.6	56.5	NA	17.0	50
Parity 3	40.8	59.5	29.5	17.0	50
Parity 4+	44.6	58.5	35.5	38.2	50

¹Average AUTO locomotion score over the 30 d preceding outcome evaluation.

²Maximum AUTO locomotion score over the 30 d preceding outcome evaluation.

³Minimum AUTO locomotion score over the 30 d preceding outcome evaluation.

⁴Percentage of AUTO locomotion score over the 30 d preceding outcome evaluation.

⁵Average locomotion score over the 7 d preceding outcome evaluation.

and apparent prevalence were calculated. Confidence intervals (95%) were calculated by bootstrap resampling at the cow level, with 10,000 bootstrap replicates drawn with replacement. For each bootstrap replicate, cows were resampled as independent units, with 1 aggregated observation per cow for each AUTO-derived metric and outcome classification. This was performed at the overall parity level, as well as separately for each parity group. Confidence intervals were calculated using the 2.5th and 97.5th percentiles of the resulting distribution for each performance and prevalence metric.

RESULTS

Descriptive Data

Initially, 522 cows were assessed during the enrollment trimmings, and 2% (11/522) were not eligible to continue due to having existing lesions (Figure 2). Additionally, 10% (54/522) of the cows were lost to follow-up, mainly due to being removed from the herd. In total, 88% (457/522) of cows remained in the study with follow-up data available after enrollment. For the validation, 99% (451/457) met the Anagnostopoulos et al. (2023) and 85% (389/457) met the Siachos et al. (2025) criteria for inclusion, with the prevalence of TT among those included being 12.6% (57/451) and 12.9% (50/389), respectively. The hoof lesions observed at follow-up for cows included

in the Anagnostopoulos et al. (2023) and the Siachos et al. (2025) evaluations are presented in Supplemental Figures S1 and S2, respectively (see Notes). Demographic characteristics of these cows are summarized in Table 3.

Performance Evaluation

At the overall parity level, the 50_BIN threshold yielded the highest accuracy and PPV compared with other classification thresholds (Figure 3). Figure 4 and Figure 5 show the classification outcomes using the threshold-based methods proposed by Anagnostopoulos et al. (2023) and Siachos et al. (2025) applied at the overall level, respectively. Only the 50_BIN threshold showed an apparent prevalence similar to the true prevalence. All other thresholds demonstrated numerically higher apparent prevalence (Figure 6).

When thresholds were applied within individual parity groups, the highest accuracy and PPV were achieved by the 50_BIN threshold applied specifically to cows in parity 2 (Figure 3). This threshold and parity group also demonstrated the greatest agreement between apparent and true prevalence (Figure 6). Parity group sample sizes varied by threshold and metric and are reported in Table 3. Across all parities and classification methods, accuracy ranged from 46.3% to 90.0%, sensitivity from 30.0% to 83.3%, specificity from 46.8% to 95.5%, PPV from 10.7% to 25.3%, and NPV from 84.6% to 96.2%.

DISCUSSION

The primary objective of this study was to assess how previously published threshold-based recommendations (Anagnostopoulos et al., 2023; Siachos et al., 2025) performed when applied to an independent dataset, by evaluating their ability to classify cows that required therapeutic intervention for hoof lesions (e.g., block or topical treatment in addition to trimming) and those that did not. Across all thresholds and parity groups, performance was characterized by relatively high accuracy and NPV but consistently low PPV, highlighting a persistent challenge in minimizing false positives. Based on work in Swartz (2025), PPV is a key metric for users. The highest PPV achieved was only 25%, limiting its practical utility, as the majority of cows flagged as positive did not have LRT. These findings suggest that the threshold approaches lack the PPV necessary to distinguish cows with hoof lesions.

Previously, Anagnostopoulos et al. (2023) reported that their 50_BIN threshold achieved high specificity (81%), NPV (84%), and accuracy (74%) but lower PPV (48%) and sensitivity (52%). In contrast, Siachos et al. (2025) found that the same threshold performed worse overall with lower accuracy (58%), specificity (76%),

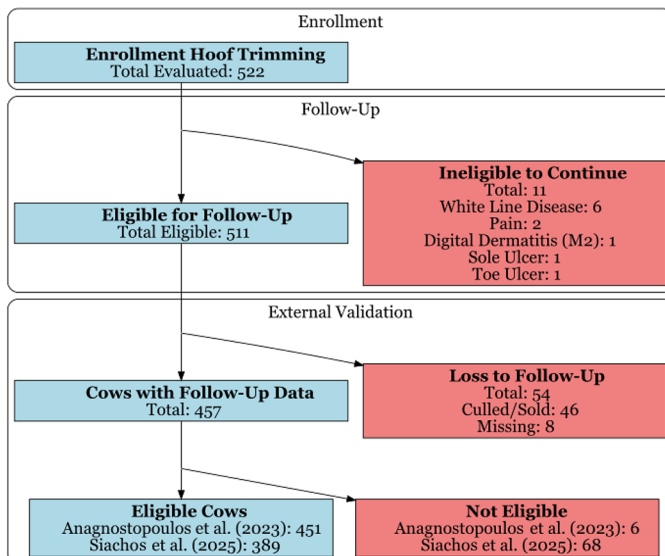


Figure 2. Flowchart illustrating the progression of cows through the study. The chart begins with cows that underwent enrollment hoof trimming, followed by those deemed eligible to continue. Of the cows eligible for follow-up, a subset was lost to follow-up. Among the cows with follow-up data, subsets were further divided based on eligibility for the criteria outlined in Anagnostopoulos et al. (2023) and Siachos et al. (2025).

Table 3. Follow-up status of animals at the outcome evaluation for the subset of cows that met the inclusion criteria outlined by Anagnostopoulos et al. (2023) and Siachos et al. (2025); table reports the overall outcome classification: lesion not requiring treatment (LNRT) $\pm$ pain, grouped as typically not treated (TNT), versus lesion requiring treatment (LRT), classified as typically treated (TT), along with lactation number at enrollment, stratified by site

Item	Anagnostopoulos et al. (2023)			Siachos et al. (2025)		
	Site 1 (n = 224)	Site 2 (n = 227)	Overall (n = 451)	Site 1 (n = 220)	Site 2 (n = 169)	Overall (n = 389)
Outcome evaluation						
LNRT	185 (82.6%)	173 (76.2%)	358 (79.4%)	183 (83.2%)	124 (73.4%)	307 (78.9%)
Pain only	14 (6.3%)	22 (9.7%)	36 (8.0%)	14 (6.4%)	18 (10.7%)	32 (8.2%)
M2-stage digital dermatitis	9 (4.0%)	8 (3.5%)	17 (3.8%)	9 (4.1%)	4 (2.4%)	13 (3.3%)
Sole ulcer	0 (0%)	4 (1.8%)	4 (0.9%)	0 (0%)	4 (2.4%)	4 (1.0%)
Toe lesion	1 (0.4%)	4 (1.8%)	5 (1.1%)	1 (0.5%)	3 (1.8%)	4 (1.0%)
White line disease	13 (5.8%)	10 (4.4%)	23 (5.1%)	11 (5.0%)	11 (6.5%)	22 (5.7%)
Other ¹	1 (0.4%)	4 (1.8%)	5 (1.1%)	1 (0.5%)	4 (2.4%)	5 (1.3%)
More than 1 type ²	1 (0.4%) ³	2 (0.9%) ⁴	3 (0.7%)	1 (0.5%) ³	1 (0.6%) ⁵	2 (0.5%)
TNT vs. TT						
TNT	199 (88.8%)	195 (85.9%)	394 (87.4%)	197 (89.5%)	142 (84.0%)	339 (87.1%)
TT	25 (11.2%)	32 (14.1%)	57 (12.6%)	23 (10.5%)	27 (16.0%)	50 (12.9%)
Lactation						
1	0 (0%)	227 (100%)	227 (50.3%)	0 (0%)	169 (100%)	169 (43.4%)
2	120 (53.6%)	0 (0%)	120 (26.6%)	118 (53.6%)	0 (0%)	118 (30.3%)
3	55 (24.6%)	0 (0%)	55 (12.2%)	54 (24.5%)	0 (0%)	54 (13.9%)
4+	49 (21.9%)	0 (0%)	49 (10.9%)	48 (21.8%)	0 (0%)	48 (12.3%)

¹Consists of different lesions reported by the site hoof trimmers: interdigital hyperplasia, periople ulcer, non-ulcer, other, puncture, vertical fissure, axial fissure, corkscrew.

²Outcomes where there was more than 1 hoof lesion.

³Consists of a white line disease and an M2.

⁴Consists of 1 white line disease and toe lesion and 1 sole ulcer and M2.

⁵Consists of 1 sole ulcer and M2.

and sensitivity (37%). Among the AUTO-derived metrics proposed by Siachos et al. (2025), the maximum LS (mMAX) threshold demonstrated greater accuracy (62%), sensitivity (41%), and specificity (77%), with PPV and NPV calculated to be 56.5% and 64.2%, respectively. Of note, these thresholds were not tested outside the study in which they were developed, and their performance was assessed using the same data on which they were originally derived, which may result in overly optimistic performance estimates and limit generalizability to other herds or management systems.

When we applied these thresholds to eligible observations in our dataset as part of the external validation, they had higher accuracy, sensitivity, and NPV, and lower specificity and PPV than the original studies. However, the low PPV highlights a limitation for practical use, as producers' perception and value depend on it (Van Nuffel et al., 2015; O'Leary et al., 2020). Low PPV can lead to an increase in the number of cows trimmed without lesions, resulting in added labor costs and short-term consequences such as reduced milk yield (Stoddard and Cramer 2024), all while incurring the ongoing expense of the detection technology. Ideally, an external validation is tested prospectively, not using a retrospective dataset as we did. Future work should evaluate its on-farm performance and economic effects in a prospective manner.

One possible explanation for the differences in performance between previous studies and our current study is the variation in enrollment criteria. Both Anagnostopoulos et al. (2023) and Siachos et al. (2025) included cows without stated eligibility criteria. In contrast, our study enrolled cows with no more than 1 prior hoof lesion, targeting animals before they became chronically lame. This design was informed by evidence showing that lameness prevalence is largely driven by chronically lame cows (Randall et al., 2018), with the goal of intervening early to prevent chronicity. This difference in study population likely contributed to the lower observed prevalence of TT hoof lesions in our study, compared with the higher prevalence estimates reported by Anagnostopoulos et al. (2023) and Siachos et al. (2025) of 31% and 42%, respectively.

Given that PPV and NPV are influenced by disease prevalence, these differences in prevalence likely affected the predictive performance metrics observed in our study. Specifically, our lower prevalence would be expected to reduce PPV and increase NPV, as lower disease prevalence typically decreases PPV and increases NPV (Tenny and Hoffman, 2025). This highlights the importance of considering population prevalence when interpreting these performance metrics. For context, Cramer et al. (2008) reported average herd-level prevalence

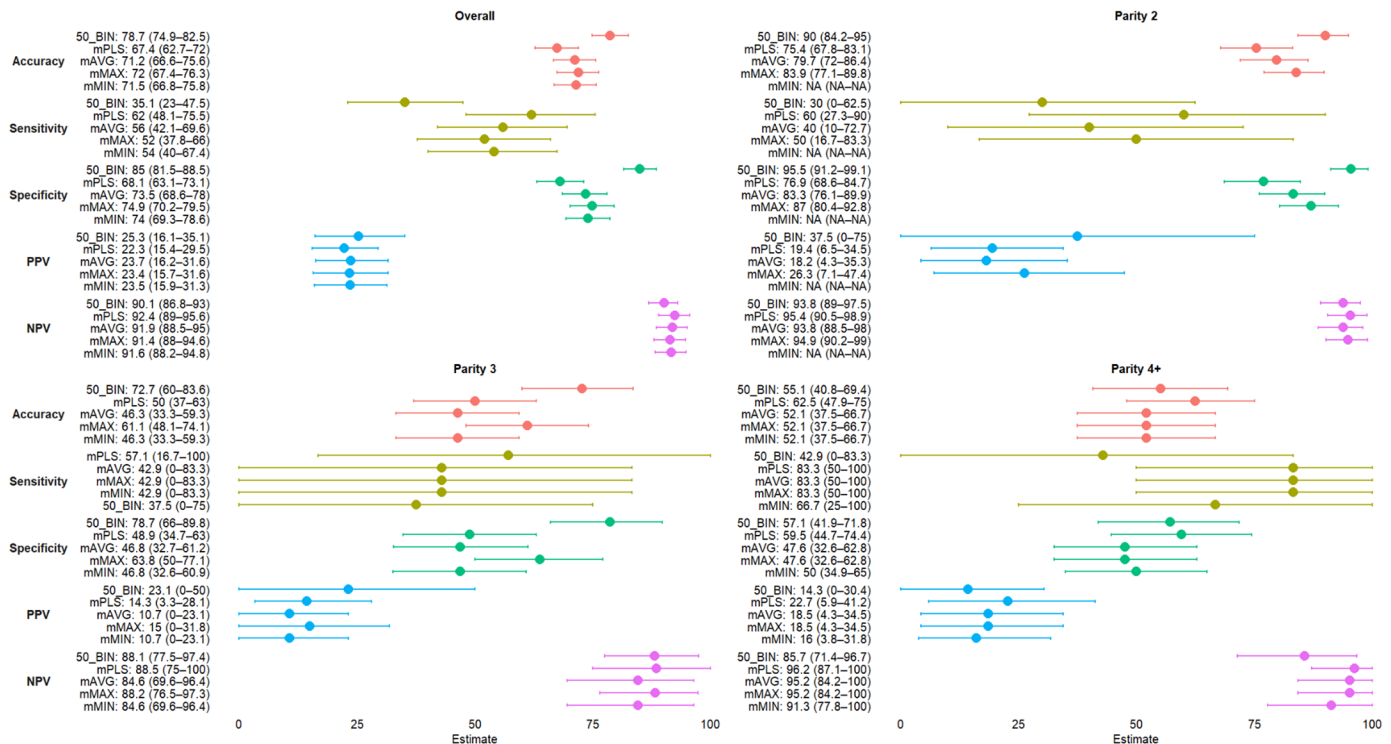


Figure 3. Performance of threshold-based locomotion score recommendations proposed by Anagnostopoulos et al. (2023) and Siachos et al. (2025), evaluated across parity groups using a subset of cows eligible under each study's criteria. Each panel corresponds to 1 parity group: overall, parity 2, parity 3, and parity 4+. Within each panel, thresholds based on different locomotion metrics are compared across 5 performance metrics: accuracy, sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV). Within each performance metric, each threshold method is presented with a metric estimate (shown as the point), with the corresponding 95% CI displayed both numerically in parentheses and visually as horizontal bars. Confidence intervals were derived from 10,000 bootstrap samples. NA indicates that the metric could not be calculated.

estimates of 24.4% for digital dermatitis, 12.5% for ulcers, 2.5% for white line abscess, and 0.4% for thin soles across 38 freestall herds, suggesting that our overall hoof lesion prevalence was lower than typically reported at the herd level.

However, our lower prevalence was an intentional design feature, aimed at detecting new hoof lesions in cows not yet chronically affected. Our observed prevalence aligns with studies focused on identifying new cases of hoof lesions, which also reported lower lesion incidence rates, including 13% when excluding corkscrew claws (Stoddard et al., 2025) and 12% when excluding corkscrews and injuries (Wynands et al., 2022). These estimates are more representative of lesion incidence in nonchronic populations. Although this approach may limit generalizability to herds with higher chronic lameness burden, it better reflects early disease detection goals, where the objective is to identify cows before the onset of chronicity.

It is also worth noting that the apparent prevalence from monthly threshold-based classifications tended to overestimate true prevalence, whereas the 50_BIN approach produced apparent prevalence estimates more

closely aligned with true prevalence. Further evaluation is needed to assess how 50_BIN performs in estimating apparent prevalence across different true prevalence scenarios. Although AUTO locomotion scores are accepted by UK Arla milk suppliers as an alternative to quarterly mobility scoring by trained observers (CattleEye, 2023), their use for estimating hoof lesion prevalence may inflate apparent prevalence by overclassifying cows as affected.

Differences in lesion classification criteria across studies likely contribute to the variations observed not only in performance metrics but also in reported prevalence. Both Anagnostopoulos et al. (2023) and Siachos et al. (2025) included sole ulcers, toe ulcers, interdigital hyperplasia, white line disease, and digital dermatitis. However, the severity thresholds applied varied, and only Siachos et al. (2025) included sole hemorrhages and axial wall fissures. In contrast, our study focused exclusively on lesions requiring therapeutic intervention and did not assess lesion severity, as our priority was to support actionable on-farm decision-making.

In our study, the primary hoof lesions targeted for detection were mainly noninfectious, specifically sole

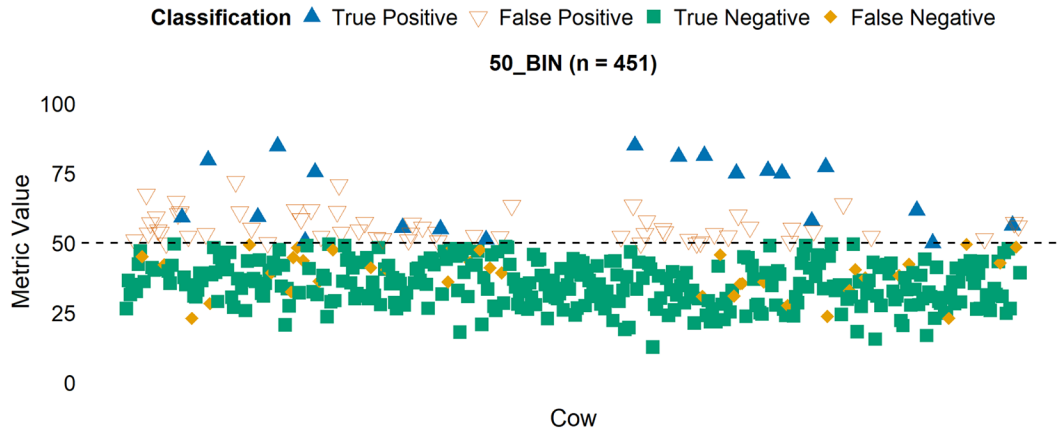


Figure 4. Predicted classification of cows into true positive, false positive, true negative, and false negative categories based on the suggested 50_BIN threshold for AUTO locomotion score metrics proposed by Anagnostopoulos et al. (2023). The 50_BIN metric represents the average locomotion score over the 7 d before follow-up evaluation. The dashed horizontal line in the panel indicates the threshold applied for classification.

ulcers, white line disease, and toe lesions that had not progressed to a chronic stage. These lesions were selected based on proposed mechanisms linking repeated injury to structural changes in the hoof. For example, Newsome et al. (2016) suggested that active lesions may trigger inflammation that stimulates abnormal bone growth at the distal phalanx, increasing pressure on the corium and leading to recurrent lesions. Similarly, Wilson et al. (2021) reported a negative association between lesion frequency and digital cushion volume, hypothesizing that chronic inflammation reduces its protective capacity. These findings supported our decision to focus on specific hoof lesions and to enroll cows

presumed not to have progressed to chronic or recurrent lesion status.

Similarly, the only infectious hoof lesion included in LRT was the M2 stage of digital dermatitis (Berry et al., 2012), as foot rot did not occur in the study population during the observation period. Some studies define only the M2 stage as active, but others also include the M1 and M4.1 stages (Cook et al., 2016). Our classification approach reflects the clinical distinction that M2 lesions are painful, respond positively to topical treatment (Solano et al., 2017), and typically necessitate individualized treatments opposed to M1 and M4.1 lesions (Bell and Vanhoudt, 2020).

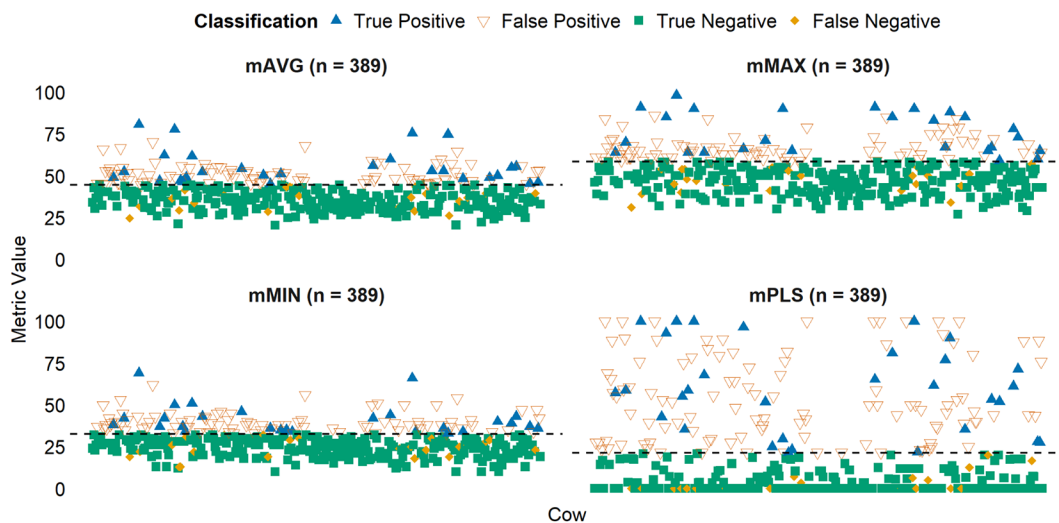


Figure 5. Predicted classification of cows into true positive, false positive, true negative, and false negative categories based on suggested thresholds for AUTO locomotion score metrics from Siachos et al. (2025). Metrics include average (mAVG), maximum (mMAX), and minimum (mMIN) locomotion scores, and the percent above 50 (mPLS), each calculated over the 30 d preceding the follow-up evaluation. The dashed horizontal line in each panel represents the classification threshold used.

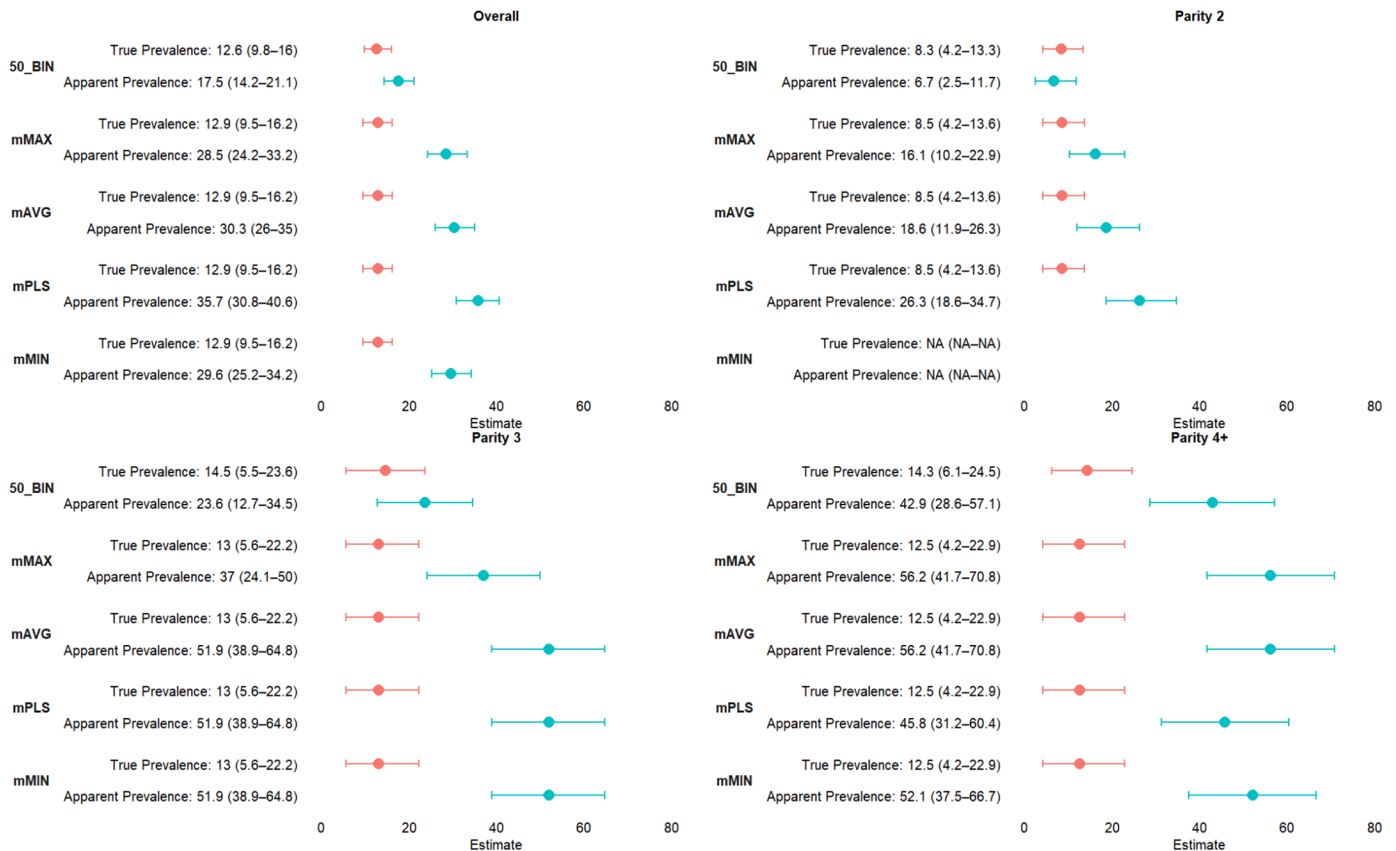


Figure 6. True and apparent prevalence of threshold-based locomotion score recommendations proposed by Anagnostopoulos et al. (2023) and Siachos et al. (2025), evaluated across parity groups using a subset of cows eligible under each study's criteria. Each panel corresponds to 1 parity group: overall, parity 2, parity 3, and parity 4+. Within each panel, different thresholding methods are shown along the y-axis. For each threshold, true prevalence (red) and apparent prevalence (blue) are presented as metric estimates (shown as points), with the corresponding 95% CI displayed numerically in parentheses and visually as horizontal bars. Confidence intervals were derived from 10,000 bootstrap samples. NA indicates that the metric could not be calculated.

Consistent with our approach for digital dermatitis, we also classified certain hoof lesions as LNRT. For example, corkscrew claws were classified as LNRT because they are associated with permanent internal structural changes and typically do not require therapeutic interventions such as blocks or topical treatments. Instead, they are managed through routine preventative trimming on a fixed schedule (e.g., every 3 mo; Cramer and Solano, 2024). Given this, excluding cows with a prior diagnosis of corkscrew claws or other lameness-causing conditions not requiring treatment, such as upper leg lameness, could enhance the system's utility by reducing the identification of cases already managed through routine trimming. However, because we did not exclude cows based on conditions identified at follow-up, our analysis provides a more practical and accurate assessment of model performance compared with the approach used by Siachos et al. (2025), who excluded cows with upper limb lameness at evaluation, regardless of hoof lesion status. Excluding cows based on follow-up findings

underrepresents the detection pool, as current thresholds cannot distinguish the cause of positive predictions, and these animals would still be eligible for prediction in real-world use. Therefore, only enrollment criteria should guide the composition of the detection pool.

In line with this classification strategy, additional lesions were also categorized as LNRT based on their limited effects on lameness or their management through routine trimming rather than therapeutic intervention. For instance, heel erosion typically results in lameness only when it extends to the corium, and The Dairyland Initiative (2025) suggests addressing it during routine hoof trimming rather than through targeted treatment. Horizontal fissures typically do not require intervention unless they extend to the corium or coronary band, and interdigital hyperplasia, when small and not interfering with the axial wall, may not cause lameness or necessitate treatment (Cramer and Solano, 2024). Additionally, although sole hemorrhages and white line hemorrhages may represent early indicators of more serious conditions

(e.g., sole ulcers or white line disease), they are not always painful and may not require immediate intervention.

One source of potential misclassification bias, and a limitation of our study, was that some cows were classified by the on-farm hoof trimmer based on farm staff identification of cows needing evaluation. Although we classified corkscrew claws, interdigital hyperplasia, and hemorrhages as LNRT, due to the automatic integration of hoof trimming software with the farm management records, they were recorded as “other” and were classified as LRT in our dataset. Specifically, the “other” category in the farm’s trimming records represents a mix of lesion types (e.g., vertical fissure, periople ulcer, puncture wounds, axial fissure, corkscrew). Although many of these lesion types were originally labeled as LNRT under our criteria, they were grouped as LRT due to the possibility that some were LRT, and the fact that these cows were flagged by farm staff, suggesting an association with lameness. Therefore, the consolidation of multiple lesion types into the “other” category may have introduced some misclassification. However, these instances represented ~1% of all observations ($n = 5$ cows).

Although this study provides insights through external validation, several limitations should be considered. First, there were few observations for evaluating the thresholds applied to the second, third, and fourth or greater lactations. This contributed to wider confidence intervals for performance metrics within these groups, reducing the precision of the estimates and highlighting the need for larger, parity-specific sample sizes in future evaluations. Additionally, lesion severity and LS were not recorded, limiting our ability to determine whether the identified lesions were associated with lameness or to assess how early in their progression they were detected. Finally, our study population primarily included cows presumed to be enrolled before lesion recurrence and chronic lameness. Taken together, these limitations influence the generalizability of our findings, as the validation set and resulting estimates may not fully represent the entire herd population or the range of lesion types, severities, and clinical presentations typically observed on-farm. Future research exploring alternative approaches, such as models that do not rely on single thresholds, may yield improved diagnostic performance for AUTO-derived metrics. This will be further discussed in Swartz et al., 2026.

CONCLUSIONS

Detecting cows with hoof lesions remains a major challenge in lameness prevention and management. In this study, we evaluated recommended metric-based thresholds to detect cows with typically treated hoof lesions. All proposed threshold-based approaches yielded high negative predictive value but poor positive predictive

value for lesions in our dataset, raising concerns about their practicality. In particular, the low positive predictive value indicates that implementation of these thresholds would likely result in unnecessary hoof trimming, with increased labor demands and economic costs. Given these findings, we do not recommend the widespread adoption of these metric-based approaches for hoof lesion detection without further refinement or evidence of financial benefit from prospective intervention studies.

NOTES

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Nonstandard abbreviations used: 50_BIN = average LS score 50 or greater; AUTO = autonomous camera-based system; LNRT = lesion not requiring treatment; LRT = lesion requiring treatment; LS = locomotion scores; mAVG = average LS; mMAX = maximum LS; mMIN = minimum LS; mPLS = percentage of LS scores

above 50; NPV = negative predictive value; P1, P2 = first and second phalanxes, respectively; PPV = positive predictive value; RFID = radio frequency identification; TNT = typically not treated; TT = typically treated.

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